

BC002 · OOP & Games

Python Bootcamp · 100 Days — Days 17–32. Tkinter, Turtle, Snake, Pong, Pomodoro.

For parents: these are the key English coding words your child meets in this course, week by week. You don't need to code — just read a word together and ask your child to explain it back in their own words. The tiny example shows what it looks like in real code.

W1

WORD	WHAT IT MEANS	TINY EXAMPLE
class	a blueprint for making objects	<pre>class Question:</pre>
attribute	a variable that belongs to an object	<pre>self.text = text</pre>
method	a function that belongs to an object	<pre>def check(self):</pre>
self	the object the method is working on	<pre>self.score += 1</pre>

W2

WORD	WHAT IT MEANS	TINY EXAMPLE
instance	one object made from a class	<pre>tim = Turtle()</pre>
method call	running a method on an object	<pre>tim.forward(100)</pre>

W3

WORD	WHAT IT MEANS	TINY EXAMPLE
object state	the current values of an object	<pre>self.x, self.y</pre>
event listener	code that waits for a key/click	<pre>screen.onkey(up, 'Up')</pre>

W4

WORD	WHAT IT MEANS	TINY EXAMPLE
inheritance	a class reusing another class's code	<pre>class Snake(Turtle):</pre>
list of objects	many objects stored in a list	<pre>self.segments = []</pre>

W5

WORD	WHAT IT MEANS	TINY EXAMPLE
collision detection	checking if two things touch	<pre>if head.distance(food) < 15:</pre>
refactor	rewrite code cleaner without changing behavior	<pre>split into methods</pre>

W6

WORD	WHAT IT MEANS	TINY EXAMPLE
scoreboard	an object that tracks and shows score	<pre>self.score = 0</pre>

W7

WORD	WHAT IT MEANS	TINY EXAMPLE
difficulty	the game getting harder over time	<pre>speed += 1</pre>

W8

WORD	WHAT IT MEANS	TINY EXAMPLE
open()	open a file to read or write	<pre>open("notes.txt", "w")</pre>
with statement	open a file and auto-close it	<pre>with open(f) as file:</pre>

W9

WORD	WHAT IT MEANS	TINY EXAMPLE
csv	comma-separated values data file	<pre>import csv</pre>

W10

WORD	WHAT IT MEANS	TINY EXAMPLE
list comprehension	build a list in one short line	<pre>[x*2 for x in nums]</pre>
zip()	pair items from two lists together	<pre>zip(letters, codes)</pre>

W11

WORD	WHAT IT MEANS	TINY EXAMPLE
tkinter	Python's built-in window/GUI toolkit	<pre>import tkinter</pre>
widget	a UI piece: label, button, entry	<pre>Button(text='Go')</pre>
command	the function a button runs when clicked	<pre>Button(command=go)</pre>

W12

WORD	WHAT IT MEANS	TINY EXAMPLE
.after()	run a function after N milliseconds	<pre>window.after(1000, tick)</pre>

W13

WORD	WHAT IT MEANS	TINY EXAMPLE
messagebox	a popup dialog window	<pre>messagebox.showinfo(...)</pre>

W14

WORD	WHAT IT MEANS	TINY EXAMPLE
try / except	catch an error so it won't crash	<pre>try: ... except: ...</pre>
json	a text format for saving structured data	<pre>json.dump(data, file)</pre>

W15

WORD	WHAT IT MEANS	TINY EXAMPLE
pandas	a library for working with table data	<pre>import pandas</pre>

W16

WORD	WHAT IT MEANS	TINY EXAMPLE
datetime	work with dates and times	<pre>datetime.now()</pre>
smtplib	send email from Python	<pre>import smtpplib</pre>